
RECURSIVE ORGANIZATION IMPROVEMENT: A MODELING SPECIFICATION FOR HUMAN-AGENT ORGANIZATIONS

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ABSTRACT

Stronger AI agents do not automatically produce better organizations: teams must also learn which work arrangements to retain and when to reconsider them. We propose a modeling specification for recursive organization improvement and evaluate it through an executable checker, a public-record mapping, and two controlled simulation studies. The specification connects actor-visible histories, organizational memory, decision rights, and evidence-carrying change contracts. Study 1 crosses six decision rules, three memory conditions, and three task environments under fixed resource ceilings. In a stationary environment, cumulative evidence raises balanced evaluation’s net value per task from 0.452 to 0.480 and nearly removes repeated reassessment’s disadvantage. A reversal of the best workflow reveals the opposite cost: indefinite retention delays adaptation, while a finite window restores eventual performance at a transition cost. In exploratory controls, matching trial acquisition and label reuse reduces the apparent reassessment gain from 0.006 to 0.002. Program replacement adds no stable benefit across the tested reversal times. Exploratory Study 2 models how human-agent review arrangements generate evidence: shared agent errors, copying, and decision rights change the preferred placement of human judgment. Paid observation can identify lower-error arrangements without recovering its cost. Both studies use synthetic parameters. The studies identify evidence generation, acquisition, reuse, and timely updating as mechanisms that must be separated from evaluator replacement when assessing organizational improvement.

Keywords human-AI interaction · multi-agent organizations · organizational learning · workflow adaptation · recursive improvement · evaluation

1 Introduction

Consider a software team that uses AI agents to draft pull requests (PRs). Faster drafting can lengthen the review queue. The team may respond by routing routine changes to automated checks and reserving human review for difficult cases. To evaluate this change, it must decide which PRs to audit, how to combine new observations with earlier evidence, and who may revise the routing or audit rule. An apparent gain can reflect a better workflow, a different sample of reviewed cases, or more effective reuse of evidence. How can the team distinguish these explanations and retain changes that improve delivery?

We propose a modeling specification for this problem and evaluate it with an executable checker, a mapping of public PR records, and two simulation studies. The running PR example connects the specification’s fields to concrete organizational decisions; the studies isolate evidence acquisition and retention, then the review arrangements that generate observable judgments.

Task-level productivity studies and studies of human-AI collaboration address different parts of this problem. Experiments on professional writing and evidence from customer support document productivity effects in particular settings [12, 41]; a meta-analysis finds substantial variation in the performance of human-AI combinations relative to their

constituent members [50]. Organization design supplies the intervening objects: the division and allocation of work, dependencies, and integration of effort [37, 45]. Complementary organizational investments can also affect the returns to a new technology [11, 39]. Consequently, comparing agent models while holding the organization fixed answers a different question from comparing organizations that can revise their workflows.

Workflow revision creates a measurement problem of its own. A new review policy changes both which cases receive attention and which failures become visible. If the same selected cases determine whether that policy is retained, organizational learning can favor a locally successful but globally harmful arrangement. An improvement procedure therefore needs an explicit account of the evidence it can acquire, the changes it can authorize, and the costs it incurs. The procedure may itself become a target of revision. Study 2 examines how review visibility and decision rights shape the evidence available for judging an arrangement.

We call this problem *recursive organization improvement* (ROI; distinct from return on investment): evaluating and retaining changes to organizational arrangements, including changes to the procedures through which those arrangements are improved. The research object is the human–agent organization. Agent capability, human judgment, coordination, evidence access, and authority can change at different rates and under different constraints.

This paper contributes a modeling specification that connects organizational state, actor-visible event histories, and evidence-carrying change contracts. Its distinctive organizing choice is to make the object of a revision and the procedure used to evaluate it jointly explicit. The specification supports reconstructing a decision from its available evidence, checking whether a patch meets declared admission conditions, and comparing retained changes under a stable outcome criterion. These are concrete modeling obligations against which an implementation can be inspected.

Study 1, *evidence acquisition and retention*, holds label quality fixed and crosses evidence-reset, cumulative, and finite-window memory with fixed evaluation programs, one-time discovery, and repeated program assessment. A template-ranking reversal makes old evidence obsolete. Matching resource ceilings and the acquisition process separates the value of remembering evidence from that of replacing an evaluator. Memory largely removes repeated assessment’s disadvantage in a stable environment, yet delays adaptation after change. Exploratory acquisition-matched controls narrow the gain attributable to evaluator replacement.

Study 2, *review arrangements and evidence generation*, derives organizational outcomes and visible judgments from individual errors, shared error sources, upstream exposure, and decision rights. Its contribution is to distinguish mechanisms that, in a two-actor case, can look identical in public records but respond differently to blinding, separate gold-labeled measurement from human override authority, and show how error dependence changes the preferred placement of humans under a fixed budget. Paid observation policies test whether identifying a better arrangement also recovers the cost of learning about it.

The argument has two strands. Study 1 follows evidence through discovery, retention, and reassessment while holding its quality fixed. Study 2 asks where observable evidence comes from: blind review preserves private judgments, whereas exposed review can replace them with copied agreement. Blinding does not remove dependence already present in private errors. The arrangement being improved therefore also determines which judgments an improvement procedure could observe. Study 2 compares fixed observation policies; it does not implement an outer search that revises those policies.

2 Conceptual foundations and modeling requirements

2.1 Organizational learning and explicit organizations

Organizational learning connects experience to retained routines [34, 38]. The distinction between the abstract understanding of a routine and its situated performance helps explain how routines can generate both stability and change [20]. Deliberate learning adds articulation and codification of experience [53], while double-loop learning addresses governing assumptions and values [4]. Engelbart’s improvement-of-improvement distinction supplies a close conceptual antecedent for examining the procedures that generate organizational change [19]. ROI translates this question into versions, events, change targets, and outcome comparisons.

Human–agent organizations also require an account of interdependence. Mixed-initiative interaction and levels of automation distinguish who initiates or performs a function [27, 42]; Coactive Design places the support for interdependent activity at the center of system design [30]. Appropriate reliance concerns when a person should use automated advice [33]. These choices alter information exposure, decision rights, and resource use, so they belong in the organizational model rather than being inferred from an actor’s capability score.

Existing formalisms already supply much of the representational machinery. MOISE+ models structural, functional, and deontic organization; OperA represents organizational interaction and agent autonomy [18, 29]. Process mining uses event data for process discovery, conformance checking, and enhancement [51]. The proposed specification con-

Table 1: Connections to established approaches. The right column identifies the records required for an ROI comparison, rather than an expressiveness ranking.

Approach	Established modeling object	Required connection for a change comparison
MOISE+; OperA	Roles, missions, norms, interactions	Target-specific authority, prior version, evidence and comparator for a revision.
Process mining	Cases, activities, event order, conformance	Organization/procedure versions, actual observation access, and sampling coverage.
Organizational learning	Experience, routines, retention	Link a retained routine to the experiment and criterion used to accept it.
Agent/workflow search	Executable candidates and evaluations	Human resources, rights, generated-output costs, and revision of the evaluator.
Generic event log	Arbitrary recorded attributes	All the same relations, if explicitly supplied; missing fields remain unknown.

nects these objects to the evaluation and retention of changes through the relations in table 1. A sufficiently annotated event log or state-transition model can encode the same relations. The contribution is their explicit organization into a comparison contract, whose completeness and execution can be checked.

Automated agent-system design and workflow optimization provide candidate-generation mechanisms [28, 52]; linguistic feedback can support behavioral revision [47]. In ROI, these mechanisms are embedded in an organization with human roles, restricted evidence, and retention decisions. The central comparison concerns organizational outcomes under specified changes, whether the candidate was written by a person, generated by an agent, or selected from a finite catalog.

2.2 From development practices to measurable arrangements

Waterfall-like staging, agile development, and Extreme Programming suggest different settings for dependencies, batch size, feedback delay, test placement, and ownership. Royce’s original account includes iteration and the risks of late system testing [46]; agile principles emphasize frequent delivery and reflection [8]; XP provides more concrete testing, pairing, and integration practices [2]. These labels identify families of configurations rather than an ordering of organizational quality. A team can retain release gates while shortening internal feedback cycles, or introduce agents into an unchanged approval pipeline.

For example, let drafting, review, and integration capacities be 4, 5, and 9 accepted-task equivalents per day. A deterministic serial capacity bound is their minimum. Doubling drafting alone raises the bound from 4 to 5; raising review capacity to 8 at the same time raises it to 8. The factorial interaction is $8 - 5 - 4 + 4 = 3$. This standard bottleneck calculation [3, 35] isolates why technical and workflow changes can be complementary. Rework, variable quality, and demand must be modeled separately in an empirical application. Our specification records the fields needed to distinguish such joint changes from an increase in task-level speed.

3 A modeling specification for recursive organization improvement

3.1 State, improvement procedure, and change boundary

An instance declares a task population, environment process, outcome criterion, and assessment horizon. The environment generates tasks and outcomes; actor policies receive only the observations assigned to them. At event index t , define

$$\mathcal{O}_t = (X_t, D_t, L_t, \Phi_t, \Pi_t, M_t), \quad U_t = (g_t, a_t, e_t, s_t, r_t). \quad (1)$$

The organization state $\mathcal{O}$ describes who does the work and what they can know or change. The procedure U describes how the team proposes, tests, selects, and retains changes. Table 2 maps their components to the PR example. Actors also have task-specific capability profiles; approval rights are recorded separately.

Definition 1 (Declared boundary). *A boundary $\mathcal{B}$ partitions fields into operational arrangements, improvement-procedure fields, and externally fixed fields for a stated comparison. It identifies authorized editors, the evaluation horizon, and the outcome criterion.*

An execution event changes work artifacts. An operational adaptation changes an arrangement under the declared improvement procedure. A procedure revision changes a field of U and evaluates its consequences for subsequent organizational changes or outcomes. For instance, assigning a PR to a reviewer is execution; replacing its routing rule is operational adaptation; replacing the sampling program used to evaluate routing changes is procedure revision.

Table 2: Core notation, illustrated by PR review routing.

Symbol	Meaning	PR example
X	Actors, roles, tasks, versioned artifacts	Reviewer, agent, PR, test report.
D	Task and resource dependencies	Approval depends on tests and reviewer availability.
L	Decision rights	Maintainer may change routing; review board may change auditing.
Φ	Observation availability	Which test results or private assessments each actor can see.
Π	Interaction and allocation protocols	Assignment order, review sequence, and workload allocation.
M	Retained evidence and versions	Audit labels, trial scores, and the adopted routing rule.
$U = (g, a, e, s, r)$	Generate, acquire, evaluate, select, retain	Propose a rule; audit PRs; score and select it; monitor continued use.
$\mathcal{B}$	Declared change boundary	Routing and auditing may change; outcome criterion stays fixed.
$\mathcal{H}_{i,t}$	Evidence visible to actor i before event t	Reports available before approval.
$p, \mathcal{T}, H$	Change contract, event trace, evaluation horizon	Recorded routing change and its follow-up period.

A fixed outer selector may govern all three. The recursive designation identifies a change target relative to $\mathcal{B}$, not a representation-independent property of software.

This boundary also resolves overlapping implementations. Φ applies the current visibility rules, while a requests evidence for a change evaluation. If one patch affects both, the implementation records the coupled fields and charges its costs once. If the outcome criterion changes, its version changes too; comparisons under the old and new criteria are reported separately.

3.2 Events and actor-visible histories

For the PR team, an event may be a reviewer assignment, a private assessment, or a revealed test report. An event z records an identifier, logical order, actor, action, input/output artifact versions, recipients, resource use, and organization version. Actions include assignment, commitment, reveal, critique, test, escalation, approval, and patch application. Logical order represents dependency; physical timestamps can additionally represent concurrency and elapsed time.

Let $\mathcal{H}_{i,t}$ contain actor i 's permitted initial information and artifacts revealed to that actor before event t . A policy satisfies

$$b_{i,t} \sim \pi_i(\cdot \mid \mathcal{H}_{i,t}, \mathcal{O}_{i,t}^{\text{vis}}), \quad (2)$$

where $\mathcal{O}_{i,t}^{\text{vis}}$ is the actor-visible projection of the organization state. Thus a simulator can score a decision using a hidden true error rate while the decision maker receives sampled labels only. A seed supports reproducibility without granting actors access to each other's random choices.

The ordering of commitment and reveal events represents human-first, agent-first, and parallel assessment. Parallel commitments preserve the opportunity for private judgments, while shared training data, common tools, or correlated task difficulty can still induce dependent errors. Statistical dependence is a joint property requiring measurements or explicit assumptions; it cannot be inferred from separate model names. Likewise, a public review record establishes that a review was published, not that another actor read it before deciding.

3.3 Change contracts and retention

A proposed routing change carries a contract: a record linking the change to its evidence, comparator, cost, and follow-up rule. Formally,

$$p = (\tau, v, \delta, \mathcal{E}, \mathcal{C}, c_p, H, \mathcal{R}_p). \quad (3)$$

τ identifies the target and proposer; v is the required input version; δ is the transformation; $\mathcal{E}$ identifies available evidence and its provenance; $\mathcal{C}$ states the comparator, population, and criterion; c_p specifies resource accounting; H is the horizon; and $\mathcal{R}_p$ specifies retention, monitoring, and rollback conditions. Contract fields may refer to separately stored artifacts. Their relationships, summarized in table 3, determine whether a comparison can be reconstructed.

Admission checks authorization, versions, available evidence, prerequisites, and resources. Efficacy is a subsequent outcome comparison. These predicates have different witnesses: permission records can justify admitting a trial, while

Table 3: Specification obligations and observable failure conditions. Unknown information is distinguished from a violated condition.

Relation	Required condition	Failure or missing record
Actor–evidence	Each decision input was available to its actor beforehand.	Hidden/future input; unknown exposure history.
Patch–state	The target version exists and matches the expected version.	Stale patch; unversioned configuration.
Actor–target	The actor holds the applicable editing right at that version.	Unauthorized change; unknown historical permission.
Evidence–population	Sampling scope and label source support the stated population.	Uncovered stratum; unknown inclusion mechanism.
Outcome–criterion	Both arms use the declared criterion and horizon.	Changed success definition; censored follow-up.
Cost–resource	Search, outputs, labels, and transitions enter one ledger.	Omitted counterfactual output; duplicate charge.
Decision–memory	Accepted and rejected proposals retain evidence and versions.	Retained policy without its validation domain.

a trial’s results can justify retaining its policy. Rejected and unsuccessful revisions remain in memory because they consume resources and influence future search.

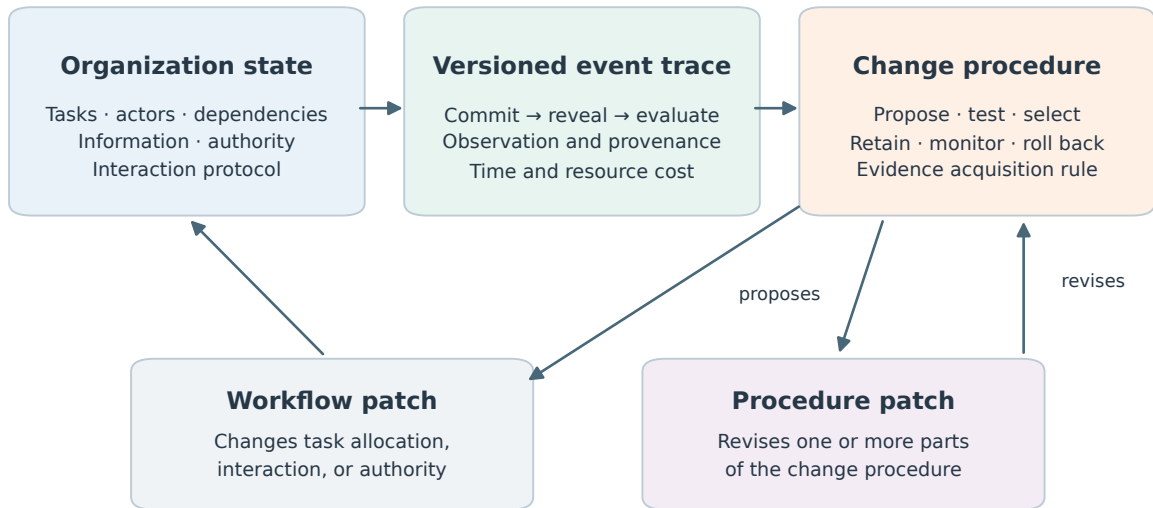
For external criterion J and specified comparator 0, the effect of a patch is

$$\Delta_H(p) = \mathbb{E}[J(\mathcal{T}_{t:t+H}^p) - J(\mathcal{T}_{t:t+H}^0)] - C_H(p). \quad (4)$$

C_H contains costs not already included in J . Randomized comparisons, controlled simulation, or explicit causal assumptions are needed to estimate the two potential outcomes. Outcomes without defensible common units remain a vector with declared constraints. In the simulations below, the net-value criteria incorporate every modeled resource cost, so no additional cost is subtracted in [equation \(4\)](#).

[Figure 1](#) connects execution to this revision loop. A software team might first change review routing, then discover that its evaluations cover only human-reviewed PRs. A second patch can alter evidence acquisition to include routine changes. That second patch is evaluated through later routing decisions and their outcomes; the mere act of adding an audit does not establish its net benefit.

A declared boundary makes the level of change inspectable



External criterion, authority limits, and evaluation horizon are declared separately.

Figure 1: The modeling specification connects organization state, event histories, and improvement procedures. A revision names its target, evidence, authority, and external evaluation criterion.

3.4 Organizational memory as a testable design choice

The memory field M separates three retained objects: workflow evidence, evaluation-program evidence, and adopted arrangements. Keeping a program name without its supporting observations preserves a decision but does not accumulate statistical knowledge. Keeping observations indefinitely can preserve knowledge about a population that no longer exists. An instance therefore records evidence provenance, applicability, and expiry alongside the retained policy.

The retention component $U.r$ specifies when evidence or arrangements are reconsidered. A fixed window is one such rule; changing the rule is a procedure revision at the declared boundary. In Study 1, memory rules are experimentally assigned and held fixed within a run. Evaluation programs may then change inside each memory condition. This factorial construction separates a design intervention on retention from the execution of a program-selection procedure.

Evidence generated during organizational search can serve more than one decision. A rejected program trial may still produce valid labels about a workflow. The trace records the labels once and reveals them to the permitted downstream evaluator; their cost is charged once. Comparing a policy that reuses those labels with one that discards them tests an information-transfer rule. Comparing two policies with the same trials and label reuse, while allowing only one to replace its program, tests a more specific contribution of program revision.

3.5 What the implementation checks

The reference implementation provides a finite checker for unique event/artifact identifiers, visible inputs, version matching, protected criteria, target-specific permissions, and patch budgets. Constructed fixtures accept a valid trace and patch and reject a hidden input and five invalid patch variants. A separate target test allows a release owner to change routing but rejects the same actor’s audit-rule change; granting the review board that right admits it. This checks the operational consequences of the declared boundary.

Study 1 instantiates a narrower executable subset: retained label statistics, retained trial scores, program versions, authorized patch application, label budgets, and a modeled cost ledger. It exports trajectory outcomes, program proportions, conditional block outcomes, and example version transitions. Full implementations would additionally need to enforce all relationships in table 3, including criterion and retention contracts. The supplied checks establish conformance to their implemented predicates; the simulation studies supply evidence about efficacy within their specified settings.

Study 2 instantiates parts of Actor–evidence through purchased private and exposed judgments, Cost–resource through acquisition, production, and switching charges, and Decision–memory through retained observation counts and recorded arrangement choices. Arrangements are selected directly from a fixed catalog; they are not submitted as versioned patches to the reference checker.

Mapping public repository records supplies a complementary feasibility check. Three merged Flask PRs yield four commit artifacts and 56 mapped events: eight reviews, 42 head-commit check runs, and six opening/merge records. Artifact identifiers and recorded actions are recoverable. Historical authority, actual message exposure, effort, organization versions, and AI involvement remain unknown. Appendix A specifies the snapshot and mapping. Such a trace can expose instrumentation gaps before a team attempts a workflow-effect estimate.

4 Study 1: Acquiring, retaining, and reconsidering evidence

4.1 Research questions and organizational instance

Study 1 follows a change from its discovery to its later use. It asks whether evidence retention alters the value of evaluator revision, whether a one-time discovery can substitute for repeated search, and which part of reassessment produces a measured gain. These questions instantiate $U.a$ (evidence acquisition), M (retained evidence), and $U.r$ (continued use or reconsideration) in the specification.

The relations in table 3 constrain specific design decisions. Decision–memory requires separate records for workflow labels, program scores, and the adopted program; we therefore intervene on these memories separately. Cost–resource requires each queried label to enter one ledger, even when both program trials and workflow selection reuse it. Actor–evidence requires current program trials to share the same pre-review information and prevents access to hidden error probabilities. Outcome–criterion keeps the deployment horizon and net-value rule common across arms. Study 1 uses these obligations to distinguish a change in the evaluator from a change in its evidence.

A proposer supplies three workflow templates, an evaluator acquires labels, and a review board may replace the evaluation program. In Study 1, the roles denote information access and decision rights and can be assigned to humans or agents; Study 2 models their individual judgments (Section 5). The environment has two equally frequent observable task strata. Error probabilities are $(.20, .20)$ for the standard template, $(.16, .16)$ for broad, and initially

Table 4: Decision rules and the comparisons they support. All arms use the same memory condition and block-level resource ceiling.

Arm	Program decision	Role in the study
Biased / Balanced / Neyman	Program specified before execution	Fixed designs with different coverage/allocation rules.
Successive rejection Discover once	Fixed elimination rule across templates Compare programs once; retain the winner	A between-template elimination comparator. Discovery followed by reuse and budget reallocation.
Repeated discovery	Reassess every eight rounds	Incremental value of continued program assessment.
Trial-matched Balanced	Run the same trials; keep Balanced	Exploratory control for evidence acquisition and timing.

(.08, .46) for specialized. Thus broad is initially best, while specialized helps one stratum and harms the population. In *stationary harm* these probabilities remain fixed. In *workflow reversal*, specialized becomes (.08, .08) before round 25 and becomes the best template. *Uniform gain* uses that improving specialized template throughout. The probabilities define synthetic mechanism contrasts, not fitted organizational parameters.

Each of 48 rounds deploys one template on 4096 production tasks. Only acquired evaluation labels enter decisions; the external scorer uses exact expected production outcomes. This separates uncertainty in organizational judgment from production sampling noise. One value unit is the gross value assigned to one completed production task before errors and costs. Net value per task subtracts error penalties and modeled expenses in these normalized units. An error costs 3 units, generation costs .02 per production or evaluation output, a label costs .20, and changing evaluation programs costs .50. Storage and computation costs are outside this ledger.

4.2 Memory and decision rules

The population-weighted posterior error estimate of template k is

$$\hat{r}_{k,t} = \frac{1}{2} \sum_{s=1}^2 \frac{1 + E_{ks,t}}{2 + N_{ks,t}}, \quad (5)$$

where E and N count observed errors and labels available to the decision. All conditions begin with Beta(1,1). *Reset* uses current-round evidence only. *Cumulative* retains all earlier evidence. *Window 8* retains evidence from the preceding eight rounds plus current observations. The same retention rule applies to past evaluation-program scores; a program’s identity remains retained until another is selected. This separates remembering a decision from remembering the evidence supporting it.

Three evaluation programs form a fixed catalog. Biased allocates 95% of a template’s allowance to the first stratum. Balanced uses equal counts. Neyman uses a two-label-per-cell pilot and posterior variance estimates to allocate the remainder across strata [40]. Each program chooses the template minimizing equation (5). Counts use population weights regardless of sampling proportions.

Workflow selection is a fixed-budget identification problem: evaluation outputs are acquired to choose a template for subsequent deployment. Successive Rejects addresses this objective by progressively eliminating candidates [5]; exploration sampling explicitly targets policy choice rather than immediate experimental outcomes [31]. We use a between-template elimination comparator based on successive rejection, whereas Neyman allocates within each template. It observes paired stratum labels, eliminates the worst estimated candidate after a first phase, and compares the remaining two after a second phase. Historical evidence enters the same estimates under the memory conditions; this is a memory-augmented implementation of the rejection schedule, with no claim to transfer stationary theoretical guarantees to the reversal environment. Allocation formulas and tie rules appear in Appendix C.

4.3 Discovery, reuse, and fixed resource ceilings

Table 4 distinguishes knowing an evaluator, discovering it once, and periodically reconsidering it. A label unit is one observed binary error outcome for one evaluated output. Every arm receives a ceiling of 2736 such observations per eight-round block. A fixed arm can allocate 342 units per round, or 114 per template for the three catalog programs. Discover once pays for program comparisons only before round 1. Repeated discovery compares programs before rounds 1, 9, 17, 25, 33, and 41. Both begin with Biased and select among the same catalog.

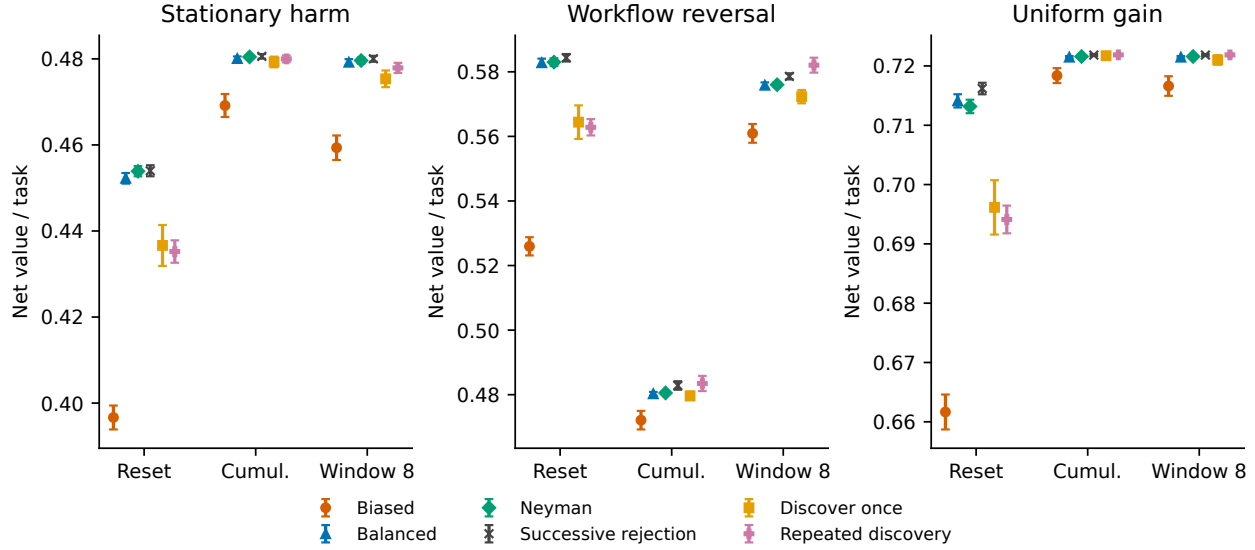


Figure 2: All core comparisons: 128 replicate trajectories per cell, means and 95% Monte Carlo intervals. Each panel uses its own value scale. “Cumul.” retains all acquired evidence; Window 8 retains the preceding eight rounds plus current evidence. The results concern the jointly specified memory, discovery, and acquisition rules.

In the base design, program trials receive at most 20% of the block ceiling, with one trial per catalog program. Each trial screens the three workflows under the same pre-review memory snapshot, selects one, and validates it on fresh balanced labels. Its score is validation value less screening expense per production task. The highest mean retained trial score determines the program. This fixed outer selector implements bounded program revision. It does not generate new algorithms.

All acquired trial labels, including evidence from rejected programs, can enter workflow memory exactly once. Decisions about program scores precede pooling the current trials; the subsequent operational screen receives the pooled evidence. This makes failed searches potential sources of reusable information. Actual trial expenditure is deducted from the current block, and the remaining allowance is spread across its operational screens. Discover once receives the full operational allowance from block 2 onward. No arm borrows from a future block; unused units due to integer allocation are not charged.

Net value includes production generation, all acquired labels, all evaluated outputs, and program changes. Selection regret measures the gross value gap to the best available template, separately from those costs. Harmful adoption counts choices worse than standard; after reversal, this measure becomes zero for all templates, so regret and adaptation trajectories are needed to distinguish them. The primary outcome averages net value over all 48 rounds, with final-eight-round performance as a separate endpoint.

The core matrix crosses three environments, three memory rules, and six arms, using 128 independent replicate streams per cell. A separate 96-replicate grid crosses program-budget shares .2, .4, .6 with trial counts 1, 3, 6, keeping the total ceiling fixed. It also removes program-score memory or trial-label pooling separately. Exploratory controls specified after inspection of the core results matched trial acquisition and tested reversals before rounds 21 and 29 on independent streams to assess attribution and timing. Appendix C specifies all conditions.

4.4 Accumulation changes the apparent cost of improvement

Figure 2 reports every core arm and memory condition. Under stationary harm, cumulative evidence raises Balanced’s mean net value from .45224 to .48007, a paired gain of .02783 (95% Monte Carlo interval .02659–.02907). Its final-eight-round value reaches .48163, the value of consistently selecting broad under its actual evaluation expenditure. Balanced’s mean selection regret falls from .02939 to .00156 per task, while its evaluation expense remains .01837. The net-value gain therefore equals the reduction in selection regret.

Repeated discovery is .01702 below Balanced with reset evidence (interval $-.01964$ to $-.01440$). With cumulative evidence, that difference becomes $-.00007$ (interval $-.00100$ to $.00085$). Discover once reaches .47928, and repeated discovery .48000. Thus the sizeable reset disadvantage does not persist when paid-for evidence is reused. This does not establish equivalence: Study 1 estimates a small difference with the stated uncertainty. Successive rejection reaches

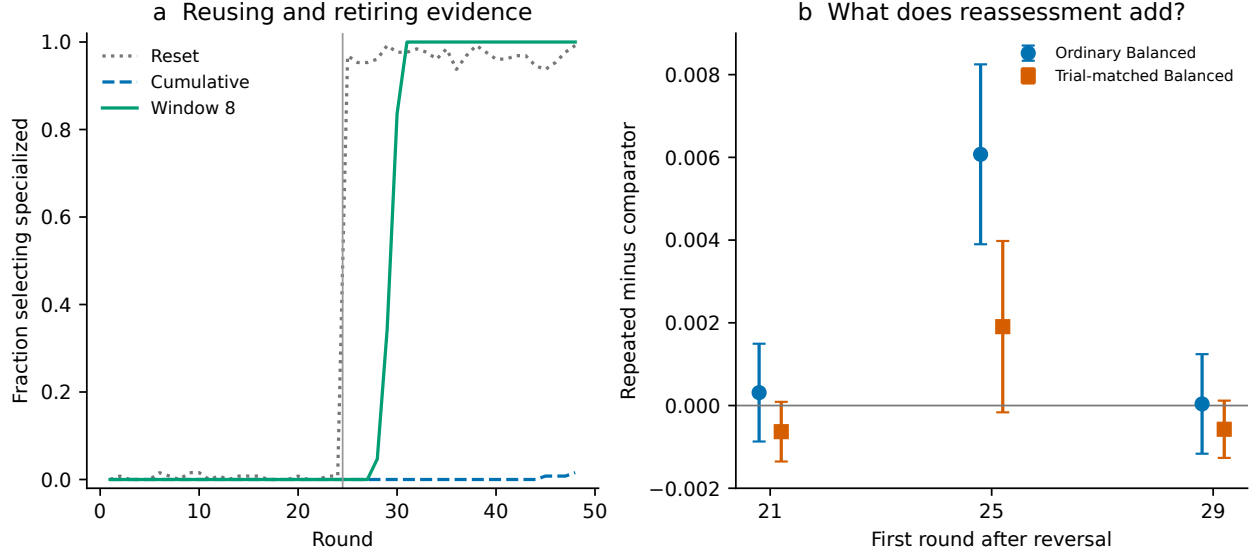


Figure 3: Retention and attribution. (a) Fixed Balanced under the core reversal; the vertical line marks the change before round 25. (b) Exploratory controls with Window 8 and reversals before rounds 21, 25, or 29. Points show repeated-discovery minus comparator differences with paired 95% Monte Carlo intervals on 128 fresh replicate streams. Matching the trials, label reuse, and timing changes the attribution of the apparent reassessment gain.

.48059 under cumulative evidence, providing a competitive fixed selection rule. Across the nine environment–memory cells, successive rejection has the largest unrounded core mean in five. In the two retained-memory Uniform gain cells, its mean is less than .00004 below repeated discovery. These are descriptive rankings, not tests of equivalence. Under Uniform gain, specialized is best from the outset, so retention improves selection without the obsolescence cost seen after reversal; a fixed elimination rule remains competitive without program replacement.

The comparison to fixed programs also depends on prior knowledge. An equal-probability initial draw from the three catalog programs defines an *ex ante* fixed-program mixture, computed as their average conditional value. Under stationary harm, its net values are .43426 with reset evidence and .47657 with cumulative evidence. Repeated discovery exceeds these prior-averaged references by .00096 and .00343, respectively. Table 6 reports the mixture in every core condition. Balanced is a prespecified representative-sampling rule, not an *ex post* oracle. These comparisons distinguish the cost of discovering a program from the value of revising an already reasonable design.

4.5 Evidence can become obsolete before an organization forgets it

Under workflow reversal, indefinite accumulation slows the replacement of broad by the newly superior specialized template. Balanced’s full-horizon values are .48026 with cumulative memory, .57583 with Window 8, and .58280 with reset evidence. The finite window improves considerably on unlimited accumulation, yet its transition delay makes it worse than reset over this particular horizon. In the final eight rounds it reaches .72163, compared with .71237 for reset and .48280 for cumulative memory. Figure 3a displays the corresponding deployment choices.

With Window 8 in the core reversal, repeated discovery exceeds discover-once by .00975 (interval .00709–.01242) and ordinary Balanced by .00623. These differences include the effects of concentrated trial acquisition and pooling, as well as the possibility of replacing the program. On the fresh exploratory streams, reversal before round 25 gives a repeated-discovery advantage of .00607 over ordinary Balanced (interval .00390–.00825). Matching program trials and label reuse while keeping Balanced reduces this difference to .00191 (interval –.00016 to .00398). For reversals before rounds 21 and 29, repeated discovery minus trial-matched Balanced is –.00063 and –.00057, respectively; both intervals span zero. The corresponding differences against ordinary Balanced are .00031 and .00004. The design therefore does not identify a stable gain from program replacement across the tested timings.

4.6 What the allocation and retention diagnostics explain

The full nine-cell budget-share/trial-count grid is reported in table 7. More trial expenditure can improve the total repeated-discovery package when it also provides reusable workflow evidence. Consequently, “meta-evaluation expenditure” is not synonymous with discarded operational evidence. Removing trial-label pooling lowers repeated

discovery from .48043 to .47751 under stationary cumulative memory, and from .58214 to .57392 under reversal with Window 8 on the independent sensitivity streams. Resetting only the program-score history has much smaller mean effects in these two conditions. These are interventions on different memory components, rather than interchangeable definitions of learning.

Table 8 reports retained-program proportions and subsequent-block outcomes conditional on retaining Balanced; Supplementary Table S2 reports all three retained programs and all three memory conditions for both discover-once and repeated-discovery rules across the three environments. At the last stationary review with cumulative memory, Biased, Balanced, and Neyman are retained in 38.3%, 31.2%, and 30.5% of runs. The 40 runs retaining Balanced have subsequent-block net value .4819 and zero harmful adoption. Accurate workflow selection therefore need not coincide with convergence to one program when programs share a well-informed workflow memory. These summaries describe selection and downstream performance together; because retained programs are selected endogenously, their conditional means are not causal program effects. The combined evidence supports a narrower and more useful explanation of organizational improvement: gains depend on what the organization learns from evaluation, how long that evidence remains relevant, and whether a measured gain survives a control with the same information acquisition.

5 Study 2: How review arrangements generate evidence

5.1 Research questions and connection to Study 1

Study 1 prescribes workflow error rates and label quality independently of the arrangement that acquires evidence. Study 2 makes the expressed judgments and final outcomes depend on a human–agent review arrangement. The same private signals may remain available under blind review or be replaced by copied agreement after exposure. Proposers and reviewers instantiate X ; exposure and order change Φ and Π ; arbitration changes L ; and observation retention instantiates M . Study 2 was designed and its protocol fixed after Study 1 was completed; it is exploratory.

Three questions organize the study. With individual accuracy fixed, how do private error dependence and exposure-induced copying separately affect decision error and visible agreement? How do these mechanisms change the preferred human–agent arrangement under a fixed budget? Which observations distinguish them, and does continuing to pay for those observations recover its cost?

5.2 Private errors, copying, and bounded review arrangements

A binary task has a proposer, agent reviewers, and, in some arrangements, a human arbiter. Humans have private error probability q_H ; conditional on task truth, their errors are independent of one another and of the agent group. Agents have private error probability q_A . With probability ρ_A , all agents on a task, including an agent proposer, share one error draw; otherwise their errors are independent. This construction preserves individual accuracy while varying private error dependence.

Motivated by observed sycophancy in multi-agent debate [43], we isolate copying (echo) without additional task information. Here η_s is the probability that an agent reviewer copies an upstream decision from source $s \in \{H, A\}$; otherwise it retains its private judgment. The subscript denotes the upstream actor, so η_H describes an agent copying a human. Blind protocols preserve private commitments for aggregation or escalation while retaining any intrinsic error dependence. We compare exposed and blind voting, exposed and blind disagreement-triggered human arbitration, random arbitration of unanimous cases, serial review, and single-actor baselines. Exact enumeration identifies a minimum-error arrangement within the declared candidate catalog and budget; expected cost breaks error ties. This error-first grid isolates decision quality within a hard resource ceiling, using cost only to break error ties. The learning comparison instead selects on deployment loss $3e + .02c$, where e is error probability and c is expected production call cost; its net-value score additionally deducts observation and switching costs (equation (12)). Appendix E specifies the joint error law, protocols, costs, parameter grid, and ties.

5.3 Visible agreement and the placement of human judgment

For an agent proposer and three agent reviewers, private accuracy of 90% and $\rho_A = .1$, increasing η_A from 0 to .8 raises unanimous agreement with the proposer from 69.06% to 91.20%, while majority-decision error rises from 3.52% to 9.64%. Conditional on an incorrect proposal, unanimous approval rises from 10.09% to 59.62%. Blind voting retains an error rate of 3.52% under the same private-signal distribution. This comparison illustrates the copying mechanism; agreement alone does not identify its cause. With one agent proposer and one equally capable agent reviewer, $(\rho_A, \eta_A) = (.8, 0)$ and $(0, .8)$ produce the same joint distribution of public votes and truth but respond differently to blinding (Appendix E).

Figure 4 separates copying from pre-existing dependence. With $q_H = .2$, $q_A = .1$, a human-to-agent call-cost ratio of four, blind overhead of .05 per reviewer, and a hard task budget of three human-call units, the minimum-error choice at $\eta_H = \eta_A = .5$ changes from an agent proposer with six blind agent reviewers at $\rho_A = 0$ to a human proposer, one blind agent reviewer, and an independent human arbiter on disagreement at $\rho_A = 1$. The budget-constrained result

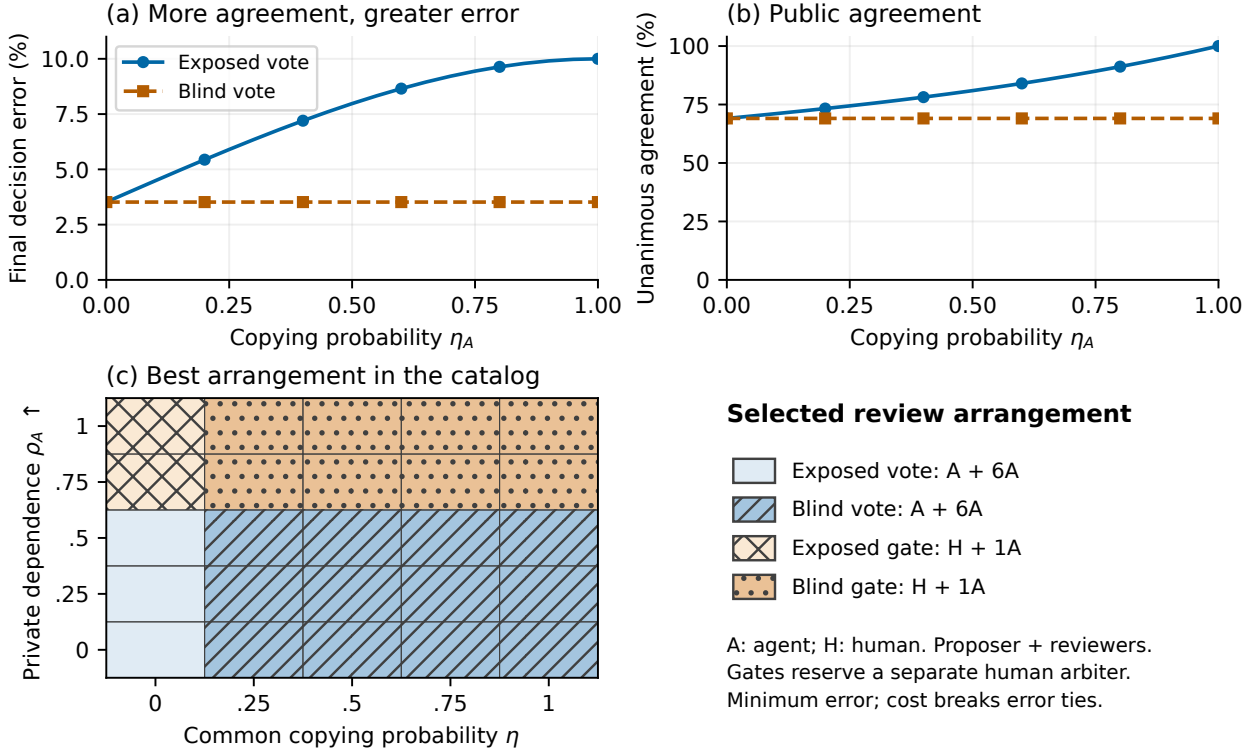


Figure 4: Study 2: private judgment, upstream exposure, and review organization. (a–b) Final majority-decision error and unanimous agreement for an agent proposer and three agent reviewers, with $q_A = .1$ and $\rho_A = .1$. Exposed reviewers copy with probability η_A ; blind protocols retain private votes. The 21 illustrative values were chosen after the core run. (c) The minimum-error arrangement, with cost breaking error ties, at each dependence and copying setting, for $q_H = .2$, $q_A = .1$, and a hard task budget of three human-call units. Human and agent calls cost 1 and .25; blind commitment adds .05 per reviewer. Gates reserve a separate human arbiter. Dependence increases upward. Colors and hatching identify the selected arrangements; all values are exact expectations.

concerns the position and authority of human judgment, not just the number of humans. A disagreement-triggered arbiter makes a final decision on selected cases. It does not provide the correct gold label assumed for learning. Under the modeled independent-arbiter rule, randomly overriding unanimous decisions helps only when the arbiter’s error probability is below the conditional error probability of those cases. Measurement and decision authority are therefore distinct design variables.

5.4 Paid observation, retention, and net value

We compare four fixed observation policies for selecting arrangements with unknown actor parameters. Across 48 rounds, they choose among 58 budget-feasible arrangements under reset, cumulative, or Window 8 memory. They acquire final outcomes only, paired private and post-exposure judgments, or observations from randomized blind and exposed tasks. An *Omit echo* ablation receives the same paired records but sets copying probabilities to zero in its predictions. Six environments, including a copying shift before round 25 and source-dependent copying, use 128 replicate trajectories per condition. All policies share a ceiling on acquisition cost, but purchase different numbers of labels. The cost schedule is separate from Study 1’s label-count budget. Observation policies are assigned treatments and do not change during learning.

In High echo with cumulative memory, paired observation achieves mean decision error 1.773% and net value .86077, compared with 9.825% and .62323 for the *Omit echo* ablation using the same acquired records. The paired net-value difference is .23755 (95% Monte Carlo half-width .00004). Randomized exposure reaches 1.782% error and

.86078 net value. The fixed blind agent-proposer/three-reviewer baseline has higher error, 3.520%, but higher net value, .87140, because it purchases no continuing learning evidence. Appendix E reports all six environments under cumulative memory; Supplementary Table S5 reports every policy–memory condition and its costs.

Across all six environments, cumulative paired observation has lower error than the fixed blind baseline, but its net value is lower in five. It spends 233.6 acquisition units per round, or .05703 per production task after amortization over 4096 tasks. In those five environments this expense exceeds the gain in deployment value, including production call cost; the switch charge is included separately. In Stronger humans, paired observation’s net value is .77306, compared with .60740 for the fixed blind baseline. The result is specific to the stated observation budget, comparator, and amortization scale.

In No echo, fixed exposed and fixed blind review have the same error rate, but net values of .87440 and .87140: blind review pays commitment overhead and one initial switch. Under cumulative memory, randomized exposure has a 56.67% optimal-choice hit rate yet the same mean error as paired observation (1.7722%); its higher production call cost makes some equally accurate choices suboptimal under deployment loss. Hit rate therefore complements error, regret, and net value rather than serving as a primary measure of decision accuracy.

Memory effects depend on the observation policy. In Echo shift, paired observation’s full-horizon net values are .85798 with reset, .86255 with cumulative memory, and .86274 with Window 8; cumulative and Window 8 both reach final-16-round error 1.7704%. For randomized exposure, the corresponding net values are .85081, .85206, and .85659, and final-16-round errors are 2.3212%, 1.8372%, and 1.7704%. Thus finite retention improves adaptation for randomized exposure in this setting, while the cumulative penalty for paired observation is small; the large reversal penalty of Study 1 does not transfer unchanged to every observation procedure.

6 Implications for human–agent organizations

6.1 Comparing capability and organizational change

The specification makes an otherwise ambiguous comparison operational. A capability-only intervention changes actor profiles while keeping assignment, visibility, authority, and improvement procedures fixed. A workflow intervention changes those arrangements under a fixed procedure. A procedure intervention changes how candidate arrangements are evaluated or retained. A factorial study can compare these interventions and their interactions under the same task population and outcome criterion. The bottleneck example in section 2 supplies one elementary prediction; Study 1 shows how evidence acquisition and retention alter the value of organizational search; Study 2 shows how review arrangements generate the judgments that can become evidence.

The distinction matters when AI changes both production volume and the work of supervision. Automation can redistribute monitoring demands [6]; human and machine learning can become organizationally interdependent [48]. In Study 1, the same amount of acquired evidence has different value depending on whether the organization reuses it, discards it, or retains it after the relevant task relationship changes. The conditional implication is to evaluate evidence retention together with the workflow that generates evidence.

The supplementary audit model in Appendix D examines the earlier decision to acquire information at all. Its finite-horizon value-of-information controller can acquire little evidence under a strong prior, while Study 1 asks what happens to evidence after it has been bought. Together the models distinguish acquisition, retention, and use. Neither an audit count nor a revision count alone measures an organization’s capacity to learn.

6.2 Independence, interaction order, and cross-functional breadth

Independent agent identities do not establish independent evidence. Shared models, training data, tools, or copied explanations can correlate errors, limiting the value of aggregation [17]. Correlated-vote extensions of Condorcet’s jury theorem analyze aggregation when votes are dependent [32]; information-cascade theory explains how observing predecessors can displace private information [9]. Social influence can also change the diversity of judgments [36]. Study 2 uses an exogenous copying probability rather than an optimizing cascade model; its contribution is the observation-equivalence comparison, the distinction between measurement and override, and the budget-dependent placement of human judgment. An ROI comparison records both pre-existing covariance assumptions and reveal events. It can then vary model diversity, information access, or communication while keeping the decision rule explicit. Section 5 instantiates this distinction by varying private error dependence and exposure-induced copying separately, while retaining the initial judgments under the blind protocols.

Human-first and agent-first interaction differ through the information available when a person commits an assessment. Private commitment can preserve an additional signal, while early advice can supply knowledge the person lacks. The outcome depends on capability, covariance, reliance, and the aggregation rule. The elementary calculation in Ap-

pendix B shows that apparent gains from exposure can equal a reweighting of private evidence. Human–AI studies of appropriate reliance and cognitive forcing motivate behavioral measurements of responses to advice [7, 13, 33]; reproducing their effects requires their task and participant conditions, not just matching a curve with selected parameters.

Cross-functional humans can reduce handoffs, interpret errors across domains, and connect agent outputs to downstream requirements. Organization design and coordination requirements identify the dependencies through which these gains could arise [15, 21, 44]. The gain must be compared with lost specialist depth, attention limits, and switching costs. A useful experiment varies human breadth while logging transfer count, rework, review delay, and quality at fixed resources. These variables are representable in the specification, while their empirical magnitudes remain task-dependent.

6.3 Weak-to-strong evaluation and more capable agents

Weak-to-strong generalization concerns learning under weaker supervision [14]. In a human–agent organization, a related difficulty appears when reviewers cannot directly assess the outputs or proposed process changes of a more capable agent. Execution capability, evaluation reliability, and authority then become separate design dimensions. The organization can invest in decomposed tests, independent evidence, selective escalation, or stronger evaluation tools; each choice changes the cost and coverage of its improvement procedure. Work on amplified supervision and scalable oversight provides candidate mechanisms for this setting [10, 16].

Hypothetical artificial superintelligence would widen the gap between execution and human evaluation capability. The same specification would require independently measurable outcomes and explicit authority for revising evaluation procedures.

6.4 Preserving disagreement and separating observation from authority

Study 2 identifies information exposure and decision rights as distinct design choices. When reviewers can copy an upstream decision, public agreement combines agreement already present in private judgments with exposure-induced conformity. An organization that escalates only visible disagreements can therefore suppress its own opportunity to correct mistakes. Retaining initial assessments, their sources, and reveal order makes these mechanisms distinguishable and supports comparisons of alternative review arrangements.

Acquiring evidence and assigning final authority also have different consequences. A gold-labeled audit informs the assessment of a workflow. A human review supplies an imperfect judgment, and granting that reviewer an unconditional override can introduce errors into cases that were already decided correctly. Under the modeled independent-arbiter rule, random overrides improve accuracy only when the arbiter’s error rate is lower than the conditional error rate of the audited cases. Human involvement should therefore be represented through capability, information access, and authority together.

6.5 How the two studies connect

The two studies connect at three points. Study 1’s workflow reversal and Study 2’s copying shift before round 25 both change the relevance of earlier observations, although their adaptation costs differ by observation policy. Study 1’s trial-matched control holds acquisition and reuse fixed while changing the retained evaluator; Study 2’s Omit echo ablation holds acquired records fixed while changing the model used to interpret them. Both comparisons show why evidence volume alone cannot explain learning outcomes. Finally, Study 1’s discover-once rule separates discovery from continued search, whereas Study 2’s paid policies keep acquiring observations throughout. A learn-then-stop comparator is needed to determine how much of Study 2’s acquisition expense can be avoided.

6.6 Validity and next empirical tests

The studies concern evaluation and review arrangements in human–agent organizations. Task decomposition, capacity, queuing, and specialist–generalist trade-offs can be represented through the specification’s task, actor, dependency, and allocation fields, but are not simulated here.

The specification is evaluated through a reference checker, a public-data mapping, and two controlled simulation studies. This establishes a path from named modeling fields to executable decisions and observable omissions. It does not yet establish lower modeling effort, better inter-rater agreement, or better predictions than alternative modeling notations. Those outcomes require users to model the same organizations under competing specifications.

For Study 1, the catalog, known stratum weights, synthetic error distributions, and cost schedule define the comparison. Program trials evaluate a fixed catalog at a smaller screening budget than operational deployment. Historical program scores average trials conducted at different knowledge states; they are a heuristic selection rule, not a stationary

posterior for program quality. Reset discards reusable evidence; cumulative retention assumes past outcomes remain relevant; the fixed window bounds age without detecting a change. These choices explain the limits of the tested learning rules. Concept-drift, switching-bandit, and change-point methods offer principled alternatives for future instances [1, 22, 23], with acquisition costs represented explicitly [49].

The acquisition-matched control shows why an improved organizational outcome should not be attributed to the most conspicuous changed component. Periodic program trials also alter when evidence arrives and who can reuse it. The observed gains do not establish that program replacement supplies a stable benefit across the tested change timings. A positive result for open-ended procedure discovery would require a broader candidate space and a comparator with equally informative observations.

Study 2 assumes conditionally independent human errors, one shared-error model for agents, and copying that adds no task information. Its parameters are experimental settings rather than estimates of human or language-model behavior. Paired observation additionally assumes that recording a private commitment does not change the subsequent copying mechanism. The enumerated optimum is specific to the candidate catalog and costs. These assumptions make exposure, dependence, and authority separately testable; their empirical magnitudes require task-specific measurement. Study 2 also holds observation procedures fixed and excludes human reviewers copying agent advice. Deployed private judgments do not enter learning memory without a separate purchase. Learning when to stop, allowing human copying, reusing deployed private judgments as feedback, and varying the observation ceiling and production amortization scale are therefore distinct next tests.

For Study 1, an organizational test would randomize review-routing changes and evaluation coverage across comparable task streams, retain failed proposals, and measure escaped errors and human attention over a declared horizon. A second stage could randomize whether teams may replace the evaluation program. Such a design would separate the value of broader evidence from the value of the procedure used to discover and retain that evidence policy.

For Study 2, an organizational test would randomize upstream visibility before private commitment across comparable review tasks, while recording actor identity, initial and final judgments, independent outcome labels, and attention costs. Separately varying escalation and override rights would distinguish the value of additional evidence from the effects of final decision authority. A later stage could randomize whether teams may replace the observation protocol, thereby testing procedure revision beyond the operational arrangement choices studied here.

7 Conclusion

Recursive organization improvement asks how human–agent teams change their work and learn whether those changes helped. The proposed modeling specification connects evidence access, organizational memory, authority, costs, and retention through explicit change contracts. Study 1 shows that evidence has a lifecycle: acquisition and reuse can remove much of repeated assessment’s apparent disadvantage, whereas retaining obsolete evidence delays adaptation, and acquisition-matched comparisons narrow the gain attributable to evaluator replacement. Study 2 identifies an earlier part of that lifecycle: review arrangements determine which private judgments become visible evidence. Shared errors, copying, cost, and decision authority jointly shape the preferred placement of human judgment, while identifying a better arrangement need not recover the cost of continued observation. Together, the studies connect how evidence is generated to how it is acquired, retained, and reconsidered. Organizational improvement must therefore be evaluated across that lifecycle, rather than inferred from agent capability or the number of revisions performed.

Code and data availability

Code, protocols, per-trajectory results, validation records, and a pseudonymized public-record snapshot are available at <https://github.com/wizardlancet/recursive-organization-improvement>. Supplementary Tables S1–S5, supplied as the arXiv ancillary file `supplementary_tables.pdf`, report Monte Carlo intervals and secondary outcomes for all conditions across both studies and the audit controls; S2 reports descriptive conditional summaries without intervals. Each table identifies its repository source, and the README maps manuscript releases and tables to source files. Code is licensed under MIT; the manuscript, figures, and synthetic data under CC BY 4.0. Third-party materials retain their original terms.

Use of AI tools

OpenAI Codex assisted with manuscript drafting and revision, simulation implementation, data analysis, and figure preparation. The two simulation studies use synthetic task outcomes and parameterized actor judgments; they do not measure deployed language-model performance or human-participant behavior.

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A Instantiating the specification: a three-PR feasibility mapping

An instance begins by declaring its population, criterion, horizon, and change boundary. It then assigns actors to roles, defines dependencies and visibility, and supplies the initial arrangement and improvement procedure. Evidence references identify versioned artifacts rather than undifferentiated transcript text. Each patch records the state it expects and the evidence its proposer could inspect. A compact contract can use the following interchange form; the referenced artifacts contain the population, protocol, scores, and retention details.

```
{
  "id": "coverage-change-1",
  "target": "evaluation.coverage_program",
  "expected_version": 0,
  "actor": "review_board",
  "transformation": {"from": "Biased", "to": "Balanced"},
  "evidence": ["program-trials-1.v1"],
  "comparison": "population-and-net-value-criterion.v1",
  "cost_ledger": "labels-outputs-transition-1.v1",
  "horizon_rounds": 8,
  "retention": "next-review-rule.v1"
}
```

This example is a proposed full contract. The executable Patch class implements target, version, actor, evidence, reversibility, and admission cost; other contract fields are supplied by the surrounding experiment protocol. Reversibility is recorded but the checker does not simulate rollback. A diagnostic coverage trace contains per-round sample counts, estimates, decisions, cost totals, and authorized program changes. The outcome equations reconstruct net value from the selected workflow and ledger, while negative fixtures exercise admission failures.

The public mapping uses a convenience-sample snapshot from `pallets/flask`: the first three merged PRs before September 30, 2026 among the first 100 closed PRs returned in descending creation order. The initial retrieval was September 28, 2026 at 17:45:43 UTC (September 29 in Singapore). Selection candidates, URLs, timestamps, and response hashes are retained. The snapshot contains factual identifiers and states, excluding message bodies and email addresses. Released actor identifiers are stable pseudonyms; account IDs and usernames are omitted. PR links remain available for provenance, so this is pseudonymization rather than irreversible anonymization.

For each PR the adapter retrieves commits, reviews, and checks at its recorded head SHA, checking pagination for the subordinate lists. Table 5 summarizes the mapped records. Commits are artifact versions; opening, review, check, and merge records are events. Endpoint definitions support these mappings [24, 25]. Earlier heads, test-merge commits, off-platform activity, and historical branch protection require additional sources. Historical permission and actual evidence

Table 5: Public-record snapshot. These counts describe record availability, not organizational improvement or AI participation.

PR	Commit artifacts	Reviews	Head checks	Events
#6133	1	0	14	16
#6096	2	5	14	21
#6095	1	3	14	19

exposure remain unknown even when a merge and a public review are visible. The adapter preserves these unknown fields and does not pass the records to the complete-trace validator. No workflow intervention or counterfactual is observed. A generic log augmented with the same missing information would support the same checks; the practical benefit tested here is identifying the required connections and missing instrumentation.

B Analytical background for interaction and evaluation

B.1 Coverage and indistinguishability

Suppose two environments induce identical distributions over every history accessible to a procedure, but a specified workflow change improves the population in one and harms it in the other. Any binary sign decision based on those histories has the same output distribution in both environments. If its probability of reporting improvement is q , its two error probabilities are $1 - q$ and q . Their sum is one, so at least one is at least $1/2$. This is the standard observation-equivalence argument.

A construction uses equally frequent flagged and unflagged cases. A change lowers flagged error from .20 to .10 in both environments; unflagged error falls to .10 in one and rises to .50 in the other. If only flagged labels are observed and no downstream signal arrives, evidence is identical although population risk changes from .20 to .10 or .30. With known positive inclusion probabilities q_i and accurate labels E_i , the Horvitz–Thompson estimator $N^{-1} \sum_i Z_i E_i / q_i$ is design-unbiased for finite-population risk [26]. Small inclusion probabilities can still produce high variance. Study 1 has positive coverage in every cell and tests this finite-evidence problem; it is not an instance of exact observational equivalence.

B.2 Dependent evidence and interaction order

For equal-variance errors with equicorrelation ρ , the variance of the mean is $\sigma^2[\rho + (1 - \rho)/n]$, with $-1/(n - 1) \leq \rho \leq 1$. Common bias contributes an additional squared-bias term to mean squared error. Information dependence therefore limits the gain from adding actors [17]; more actors alone need not produce proportionally more information.

Let independent initial human and agent estimation errors have variances v_H and v_A . Suppose the human sees the agent answer and replaces the private judgment by $(1 - \alpha)H + \alpha A$. Equal averaging of that judgment with A has risk

$$R(\alpha) = \{(1 - \alpha)^2 v_H + (1 + \alpha)^2 v_A\} / 4. \quad (6)$$

For $v_A < v_H$ and $0 \leq \alpha \leq 1$, the optimum is $\alpha^* = (v_H - v_A) / (v_H + v_A)$, giving $v_H v_A / (v_H + v_A)$. This is also the risk of optimal inverse-variance weighting of the original private judgments. Thus a gain over equal private averaging is a reweighting gain, not evidence that exposure creates information. With initial covariance σ_{HA} , risk gains the term $(1 - \alpha^2)\sigma_{HA}/2$. The parameters describe a candidate interaction model; they are not empirical estimates of anchoring.

Cross-functional breadth can be represented by reduced transfer cost and changed capability profiles. A simple accounting decomposition is saved handoff cost plus improved integration, minus lost specialist depth and switching cost. Its sign depends on measured magnitudes and on the dependency graph. The specification preserves these separate terms instead of assigning a universal benefit to generalists.

C Study 1: complete specification

C.1 Core protocol and exploratory controls

The core protocol, `examples/learning_protocol.json`, records a design fixed before execution and is archived at commit 75d312e. This first public commit contains both protocol and results; it does not establish a pre-run Git timestamp. The protocol specifies 48 rounds, a 2736-label ceiling for each eight-round block, three catalog programs, six arms, three memory conditions, three environments, and the full allocation grid. There was no parameter selection on test results. The trial-matched control and change-timing checks were specified after inspection of the core results to examine a possible acquisition-timing explanation. Their design and independent random streams are described below. They use fresh streams and are reported as exploratory attribution checks. Neither protocol is an external preregistration.

The core has $3 \times 3 \times 6 \times 128 = 6912$ trajectories. The allocation grid uses 96 replicates for each share/trial-count combination and each discovery arm under stationary cumulative memory and reversal with Window 8. Four fixed references and two memory ablations in each environment bring this analysis to 4608 trajectories. The exploratory controls cross four environments (stationary and reversals before rounds 21, 25, 29), three memory conditions, four arms, and 128 replicates: 6144 trajectories. The three independent keyed root seeds are 920000, 930000, and 940000. Each random stream is indexed by root seed, round, acquisition stage, program, trial, and replicate. Conditions share underlying potential-label streams within each root; no policy receives unqueried labels.

C.2 Evidence update and program trials

For a template–stratum cell, cumulative memory adds each actually queried label once. Window 8 uses observations at historical rounds $t - 8$ through $t - 1$, plus current observations. Reset retains no earlier observations. All program trials at a review receive the same pre-review memory snapshot. Trial outputs and fresh validation labels are pooled only after program selection, then supplied to the current operational screen. Validation observes only the selected template, while screening observes all templates except for the second phase of the rejection comparator. A label from a rejected program remains usable if its source and population are appropriate. The pooling ablation withholds all program-trial labels from workflow memory while retaining program scores.

Each trial score is $1 - 3\hat{r}^{\text{val}} - .22N^{\text{screen}}/4096$, where validation uses equal counts across the two strata. A program’s score is the mean of its trial means retained under the memory rule. Each review has the same trial count within a configuration. Ties favor Biased, then Balanced, then Neyman. Program-score memory can be disabled independently

of workflow memory. Retained scores were generated under different historical knowledge states; they are empirical assessments used by this specified controller, not identically distributed observations of an invariant program value.

For block budget $B = 2736$, program share f , three programs, and q trials per program, a trial receives $b = \lfloor \lfloor fB \rfloor / (3q) \rfloor$ label units. Validation obtains $v = \max\{1, \lfloor .2b \rfloor / 2\}$ labels per stratum; screening receives $b - 2v$ total units. The base $(f, q) = (.2, 1)$ therefore allows 182 units per trial, with 18 validation labels per stratum and 146 units for screening. Actual counts can be smaller after integer allocation. Let M_i be the actual number of trial labels for replicate i at a block's start. Its subsequent per-round operational allowance is $\lfloor (B - M_i) / 8 \rfloor$. Discover once has $M_i = 0$ after block 1. This rule keeps the budget fixed when trial count or program share changes.

C.3 Within-template and between-template allocation

For the catalog programs, let u be one template's share of the operational or trial allowance. Balanced obtains $\lfloor u/2 \rfloor$ labels in each stratum. Biased obtains $\max\{1, \lfloor .95u \rfloor\}$ and $\max\{1, \lfloor .05u \rfloor\}$. All tested allocations fit the allowance. Neyman first obtains $m = \min\{2, \max\{1, \lfloor u/4 \rfloor\}\}$ labels per cell. Using retained and pilot data, it estimates p_s , sets $v_s = p_s(1 - p_s)$, and allocates

$$n_s = m + \left\lfloor \frac{(u - 2m)\sqrt{v_s}}{\sum_{\ell} \sqrt{v_{\ell}}} \right\rfloor. \quad (7)$$

Pilot labels enter the posterior once. Equal label costs make this the equal-cost stratified allocation. Catalog selection breaks posterior ties in template order standard, specialized, broad.

The successive-rejection comparator treats one pair of stratum labels as a bounded loss with mean equal to population risk. With n affordable pairs and $K = 3$, it uses the schedule of Audibert et al. [5]:

$$\overline{\log K} = \frac{1}{2} + \sum_{j=2}^K \frac{1}{j}, \quad n_k = \left\lceil \frac{n - K}{(\log K)(K + 1 - k)} \right\rceil, \quad k = 1, 2. \quad (8)$$

All three templates receive n_1 pairs; the worst posterior-mean template is removed, and the remaining two reach n_2 pairs each. Elimination ties remove the lowest indexed template; final ties use the lowest surviving index. Historical evidence enters the posterior under the memory conditions. Thus the implementation uses the published rejection schedule with declared posterior ranking and deterministic ties. At the fixed allowance of 342 labels it uses 336 labels. No theoretical bound is claimed for this modification under memory or drift. SR is a fixed between-template comparator and is not part of the three-program discovery catalog.

C.4 Outcome accounting, checks, and uncertainty

For deployed template k_t with true population risk $r_{k_t, t}$, total queried labels Q_t , and program-change indicator S_t , the external score is

$$V_t = 1 - 3r_{k_t, t} - .02 - \{.22Q_t + .50S_t\} / 4096. \quad (9)$$

Each queried label corresponds to a generated evaluation output, so .22 includes .20 labeling and .02 generation. The separate .02 term prices each production output. Unqueried potential-label arrays are a simulator device, not actor-visible generated outputs. All rejected-program trials are charged. Program comparison costs are not subtracted a second time after computing net value.

The simulator checks every block ceiling and the identity $V_t = 1 - 3r_t^* - .02 - \text{regret}_t - \text{expense}_t$, where r_t^* is the best available risk. An independent scalar reconstruction checks 36 fixed-Balanced trajectories across the three environments and memory conditions to absolute tolerance 10^{-12} . Separate fixtures check the lookback boundary, rejection allocation, and selection of a known best template. The patch checker enforces target authority and state version for program changes.

Reported half-widths are $1.96s/\sqrt{N}$ across independent replicate trajectories. Paired intervals use within-replicate differences under the common streams. They quantify simulation uncertainty conditional on the supplied parameters. The uniform fixed benchmark averages the conditional mean values of Biased, Balanced, and Neyman under an initial uniform program prior; its paired contrast integrates over this discrete prior rather than adding random program draws. No claim of equivalence or correction for multiple hypothesis testing is attached to intervals spanning zero.

Table 6 reports all core net-value means. Supplementary Table S1 reports their Monte Carlo intervals, costs, harmful adoption, selection regret, and late outcomes; replicate-level data accompany the reproducibility materials.

Table 7 reports the complete factorial grid, so favorable settings are not selected for the main result. Supplementary Table S2 reports retained-program proportions, conditional sample sizes, and net value and harmful adoption over the subsequent eight rounds. These conditional means remain descriptive because the retained program is selected using noisy evidence.

Table 6: Complete core net-value means. B: Biased; Bal: Balanced; N: Neyman; Mix: uniform fixed-program mixture; SR: successive rejection; O: discover once; R: repeated discovery. Each cell uses 128 trajectories. Monte Carlo intervals and additional outcomes are reported in Supplementary Table S1.

Environment	Memory	B	Bal	N	SR	O	R	Mix
Stationary harm	Reset	0.3967	0.4522	0.4539	0.4540	0.4366	0.4352	0.4343
Stationary harm	Cumulative	0.4692	0.4801	0.4805	0.4806	0.4793	0.4800	0.4766
Stationary harm	Window 8	0.4594	0.4792	0.4796	0.4800	0.4754	0.4779	0.4727
Workflow reversal	Reset	0.5260	0.5828	0.5830	0.5844	0.5644	0.5628	0.5639
Workflow reversal	Cumulative	0.4721	0.4803	0.4805	0.4829	0.4796	0.4835	0.4776
Workflow reversal	Window 8	0.5609	0.5758	0.5760	0.5786	0.5723	0.5821	0.5709
Uniform gain	Reset	0.6617	0.7141	0.7132	0.7162	0.6962	0.6941	0.6963
Uniform gain	Cumulative	0.7184	0.7215	0.7216	0.7218	0.7217	0.7219	0.7205
Uniform gain	Window 8	0.7166	0.7215	0.7216	0.7218	0.7210	0.7219	0.7199

Table 7: Repeated discovery minus discover-once at fixed total budget. All program shares and trial counts are shown. Entries are paired differences $\pm$ 95% Monte Carlo half-width; 96 fresh replicates per condition.

Program share	Trials/program	Stationary, cumulative	Reversal, Window 8
0.2	1	+0.0012 $\pm$ 0.0012	+0.0071 $\pm$ 0.0024
0.2	3	+0.0019 $\pm$ 0.0020	+0.0053 $\pm$ 0.0023
0.2	6	+0.0006 $\pm$ 0.0008	+0.0062 $\pm$ 0.0020
0.4	1	+0.0002 $\pm$ 0.0004	+0.0105 $\pm$ 0.0025
0.4	3	+0.0002 $\pm$ 0.0001	+0.0098 $\pm$ 0.0026
0.4	6	+0.0002 $\pm$ 0.0001	+0.0080 $\pm$ 0.0019
0.6	1	+0.0007 $\pm$ 0.0011	+0.0175 $\pm$ 0.0021
0.6	3	+0.0001 $\pm$ 0.0001	+0.0158 $\pm$ 0.0022
0.6	6	+0.0002 $\pm$ 0.0002	+0.0134 $\pm$ 0.0015

Table 8: Repeated discovery: retained-program percentages and subsequent eight-round outcomes conditional on retaining Balanced. Stationary and uniform gain use cumulative memory; reversal uses Window 8. B/Bal/N are Biased/Balanced/Neyman; n_{Bal} is the number retaining Balanced among 128 replicates. Harm is the percentage of rounds deploying a template worse than standard. Supplementary Table S2 reports all three retained programs and all three memory conditions for both discover-once and repeated-discovery rules across the three environments.

Environment	Review	B (%)	Bal (%)	N (%)	n_{Bal}	Net given Bal	Harm (%)
Stationary	1	27.3	37.5	35.2	48	0.4747	0.00
Stationary	9	26.6	35.9	37.5	46	0.4819	0.00
Stationary	17	33.6	34.4	32.0	44	0.4819	0.00
Stationary	25	33.6	35.9	30.5	46	0.4819	0.00
Stationary	33	35.9	32.0	32.0	41	0.4819	0.00
Stationary	41	38.3	31.2	30.5	40	0.4819	0.00
Reversal	1	27.3	37.5	35.2	48	0.4747	0.00
Reversal	9	26.6	35.9	37.5	46	0.4783	0.00
Reversal	17	43.8	27.3	28.9	35	0.4819	0.00
Reversal	25	39.8	30.5	29.7	39	0.6188	0.00
Reversal	33	43.8	28.1	28.1	36	0.7219	0.00
Reversal	41	47.7	20.3	32.0	26	0.7219	0.00
Uniform gain	1	35.9	27.3	36.7	35	0.7219	0.00
Uniform gain	9	26.6	30.5	43.0	39	0.7219	0.00
Uniform gain	17	25.8	29.7	44.5	38	0.7219	0.00
Uniform gain	25	28.1	30.5	41.4	39	0.7219	0.00
Uniform gain	33	28.1	28.1	43.8	36	0.7219	0.00
Uniform gain	41	30.5	25.8	43.8	33	0.7219	0.00

D Supplementary audit-control model

The supplementary audit model isolates longitudinal evidence acquisition under a fixed controller. It contains 160 periods of 32 tasks. Route A has base value 1 and error probability .04 (stationary) or alternating .04/.25 blocks of 40 periods; route B has base value .82 and error probability .08. An error costs 3, an audit costs .20, and an audit-rate change costs .50 per batch. Only audits reveal A’s errors, available next period; executing B can still be paired with acquiring an A audit. Observations update a sliding-window Beta estimate. The base prior is Beta(1, 9) and window 12.

The payoff-optimal routing threshold is .14, since $1 - 3p = .82 - 3(.08) = .58$. The heuristic adaptive controller audits at rate .40 when posterior $P(p > .14)$ lies between .10 and .90, and at .025 otherwise. Periodic dense auditing uses .40 for the first three periods of each 12-period cycle and .025 otherwise. The base audit comparison uses 400 seeds from 20260928, common across policies. Stationary net values are .86089 for fixed .025, .86395 for fixed .05, .85841 for fixed .10, and .87976 for no auditing. No auditing retains A under the base prior, the optimal route in that stationary case; audits cannot repair current outputs.

With slow shifts, adaptive auditing exceeds fixed .10 by .00857 per task (paired interval .00676–.01038). A separate 200-seed grid, beginning at 20360928, crosses windows 6, 12, 24 with stationary, 40-period, and 10-period regimes. It includes no audit, fixed .05/.10/.40, periodic dense, and adaptive policies. Supplementary Table S3 reports all 54 combinations of environment, memory window, and audit policy.

D.1 Finite-horizon EVSI controller

The finite-horizon expected-value-of-sample-information (EVSI) controller evaluates acquiring $m \in \{0, 1, 4, 16\}$ labels. Given current Beta parameters (a, b) , a beta-binomial predictive distribution over k observed errors gives posterior mean $(a + k)/(a + b + m)$. Let $L_\theta(p)$ be $1 - 3p$ if $p \leq \theta$ and .58 otherwise. The controller maximizes

$$h\{\mathbb{E}_k[L_\theta((a + k)/(a + b + m))] - L_\theta(a/(a + b))\} - \frac{[(.20 + g_c I_B)m + .50 I_{\text{change}}]}{32}. \quad (10)$$

I_{change} equals 1 when the audit rate changes and 0 otherwise. Both costs in the bracket are subtracted. I_B indicates executing B; g_c is the cost of generating a counterfactual A output to audit. The predictive sum is exact under the current Beta model. The controller approximates future value with horizon h and a sliding window, so it is a finite-horizon EVSI heuristic under drift.

Development fixes the prior at Beta(1, 9), the threshold at .14, and $g_c = 0$. Seeds 710000–710063 choose h from 1, 4, 8, 16 using stationary and slow-shift mean value; $h = 16$ is selected. Testing uses 96 independent seeds 810000–810095, three priors, thresholds .14/.16, generation costs $g_c = 0/.20$, two environments, and three policies: 6912 trajectories. Threshold .16 is a misspecified routing-rule sensitivity under unchanged payoffs. Nonzero g_c charges selectively generated A outputs while B is executed; it is distinct from a common cost charged for producing every candidate under every policy.

Table 9 reports all prior/threshold means at $g_c = 0$. Supplementary Table S4 reports both candidate-generation cost conditions, marginal Monte Carlo intervals, and paired policy contrasts. EVSI’s performance depends strongly on the prior: with no initial feedback, information acquisition itself can stall. Four regression cases match the preserved scalar implementation to 10^{-12} .

Table 9: Supplementary audit sensitivity at zero additional candidate-generation cost. Columns give net-value means over 96 trajectories; F: fixed .10, A: adaptive heuristic, E: EVSI. Uncertainty estimates and results at $g_c = .20$ are reported in Supplementary Table S4.

Prior ($a : b$)	Threshold	Stationary			Slow shifts		
		F	A	E	F	A	E
1:1	0.14	0.8479	0.8530	0.8338	0.6616	0.6753	0.6758
1:1	0.16	0.8523	0.8571	0.8342	0.6597	0.6729	0.6757
1:9	0.14	0.8581	0.8649	0.8656	0.6620	0.6706	0.6599
1:9	0.16	0.8592	0.8686	0.8660	0.6515	0.6642	0.6526
1:19	0.14	0.8595	0.8720	0.8795	0.6468	0.6571	0.5673
1:19	0.16	0.8595	0.8735	0.8795	0.6263	0.6396	0.5673

E Study 2: complete specification

E.1 Private judgments, exposure, and decision rights

Protocol provenance. Study 2 was designed and its protocol fixed after Study 1 was completed; it is exploratory. The protocol is `study2/protocol.json` (archive commit `fdf5ec1`). Its retained pre-run record, `study2/qa/pre_run_record.json`, records a matching SHA-256 hash at 13:06:04 UTC on October 2, 2026. The Git archive postdates execution; the local hash record is not an independently timestamped preregistration.

A task has balanced binary truth $y \in \{0, 1\}$. A private judgment equals y unless its actor makes an error. Human errors are independent Bernoulli draws with probability q_H , conditional on y , and independent of the agent group. For each task, draw a shared-mode indicator with probability ρ_A . In shared mode, all agent errors equal one Bernoulli draw with probability q_A ; otherwise each agent draws an independent Bernoulli error with that probability. Tasks are independent. An agent proposer belongs to the same group as its reviewers. This mixture preserves marginal agent error q_A and gives pairwise error correlation ρ_A ; the full mixture law, rather than correlation alone, determines collective error.

Let d_0 be the displayed upstream judgment and d_i^{priv} reviewer i 's private judgment. An exposed reviewer reports

$$d_i^{\text{pub}} = (1 - C_i)d_i^{\text{priv}} + C_id_0, \quad C_i \sim \text{Bernoulli}(\eta_s), \quad s \in \{H, A\}. \quad (11)$$

Here η_s is the probability that an agent reviewer copies an upstream decision from source s ; s identifies a human or agent upstream actor. Copy indicators are independent of private errors and of one another. This binary rule selects one judgment; it does not average signals or create new task evidence. Blind protocols use private commitments for aggregation or escalation, so changing η_s does not change their decisions. In serial review, the immediately preceding review replaces d_0 at each step, and all sources after the first step are agents.

Table 10 defines the six protocol families and two single-actor baselines. A gate retains the proposal only if all relevant reviews agree with it; otherwise a separate human makes the final decision. The human arbiter has error probability q_H and is independent of the proposer and reviewers. A human proposer and a human arbiter therefore require two distinct humans. A production audit grants the arbiter override authority, whereas a gold label supplies correct measurement for learning. These rules separate private information, public disagreement, and final authority. For example, the

Table 10: Review protocols and final decision rights. Voting includes the proposer and all reviewers; a tied vote uses a fair random choice. All reviewers are agents. Gates call an additional human only on the indicated cases.

Protocol	Evidence used	Final decision
Exposed vote	Reviews after seeing the proposal	Majority vote.
Blind vote	Private commitments	Majority vote.
Exposed disagreement gate	Public reviews versus proposal	Human arbitration on any disagreement.
Blind disagreement gate	Private reviews versus proposal	Human arbitration on any disagreement.
Gate with random arbitration	Public reviews versus proposal	Arbitrate all disagreements and a random 25% of unanimous cases.
Serial last-reviewer decision	Each reviewer sees only its immediate predecessor	Last review; no collective vote.
Single actor	One human or agent private judgment	That actor's judgment.

change in error from randomly overriding a unanimous case is $q_H - P(d_0 \neq y \mid \text{unanimous})$. An independent human override helps only when this difference is negative.

The exact search crosses $q_H \in \{.1, .2, .3\}$, $q_A \in \{.05, .1, .2, .3\}$, $\rho_A \in \{0, .25, .5, .75, 1\}$ and $\eta \in \{0, .25, .5, .75, 1\}$ with $\eta_H = \eta_A = \eta$, human-to-agent call-cost ratios 2, 4, 8, blind overheads 0, .05, .2 per reviewer, task budgets 1, 2, 3, and hard versus expected budget constraints. Human calls define one cost unit. A hard budget reserves every potentially called actor; an expected budget constrains the mean cost including escalation probabilities. The catalog has $2 \times 7 \times 6 + 2 = 86$ arrangements: two proposer types, one to seven reviewers, six protocols, and two single-actor baselines. Exact finite-state probabilities identify the minimum-error feasible arrangement in each of 16,200 cells. Errors within 10^{-12} count as ties; lower expected cost breaks error ties. Remaining ties use catalog order: human then agent baseline, then human/agent proposer, increasing reviewer count, and the six protocols in table order. The deployment-loss optimum is recorded separately.

The fixed-roster comparison retains all 25,200 combinations of the accuracy, dependence and exposure grid, two proposer types, one to seven reviewers, and six protocols. A separate source-sensitivity grid uses $\eta_H, \eta_A \in \{0, .2, .5, .8, 1\}$, $\rho_A \in \{0, .5, .9\}$, $q_H = .2$, and $q_A = .1$, giving 6,450 arrangement evaluations. The 21 equally

spaced exposure values in figure 4a–b are exact illustrative evaluations chosen after the core run, not a prespecified test.

E.2 Paid observations and retained evidence

An observer can learn individual ability and copying only from the information it buys. For an agent proposer and one equally capable agent reviewer,

$$P(d_1^{\text{pub}} \neq d_0) = 2q_A(1 - q_A)(1 - \rho_A)(1 - \eta_A).$$

Consequently, at the same q_A , $(\rho_A, \eta_A) = (.8, 0)$ and $(0, .8)$ give the same joint distribution of public votes and truth in this two-actor case, but different responses to blinding. Private observations or randomized visibility can distinguish the mechanisms.

Four fixed observation policies select among 58 arrangements with at most five reviewers and hard task cost at most three. Human, agent, gold-label, and per-reviewer blind-commitment costs are 1, .25, 1, .05, respectively. Each round permits at most 240 acquisition-cost units, hence 1920 per eight-round block, with no borrowing. These cost units differ from the label counts used in Appendix C. All current observations are acquired and charged before the selected arrangement is deployed; production outcomes do not provide free learning feedback.

Paired observation records two human and six agent private judgments, ten additional exposed agent responses (five for each proposer type), a gold label, and six private-commitment overheads on each task. Its cost is $2 + 6(.25) + 10(.05) + 1 + 6(.05) = 7.3$, allowing 32 tasks and costing 233.6 per round. The same task supplies pre- and post-exposure judgments. Among reviews that initially disagree with a source, the fraction switching to that source estimates η_s . Study 2 assumes recording the private commitment does not itself change the later copying mechanism.

Randomized exposure acquires 50 tasks, randomized in balanced cells crossing human/agent source and blind/exposed review. Each task purchases two human responses, six agent responses, and a gold label. Blind tasks additionally pay six commitment overheads. The four cells have 12 or 13 tasks, with 25 blind tasks in total, costing $25(4.8) + 25(4.5) = 232.5$ per round. Private ability and dependence use only blind tasks. For each source, the estimate of copying is one minus exposed disagreement divided by blind disagreement, clipped to $[0, 1]$. This design uses disjoint blind and exposed tasks and assumes exchangeability across their randomized assignments.

Both model-based policies estimate q_H and q_A from gold-labeled private errors with one error and one success pseudocount, then clip estimates to $[\text{.001}, \text{.499}]$. Agent dependence is the private pairwise error moment minus squared raw error frequency, divided by the raw Bernoulli variance and clipped to $[0, 1]$; zero estimated variance gives zero dependence. Paired copying estimates use half an event in each outcome as pseudocounts. Randomized disagreement rates likewise use half-event pseudocounts before taking their ratio. Each policy pools the requisite counts over its assigned memory interval and predicts every feasible arrangement’s error and call cost under the known model family. The true parameters are used only by the external scorer and catalog benchmark.

Omit echo ablation purchases exactly the paired observations but fixes $\eta_H = \eta_A = 0$ when predicting arrangements. *Final-outcome sampling* instead buys an arrangement’s executed final outcome and a gold label, cycling through candidates. At zero-based round r , trial j uses catalog index $(83r + j) \bmod 58$. A trial reserves its worst possible call cost plus one gold label; actual calls and that label are charged. It stops before exceeding the ceiling. For a candidate with n retained observations, e errors, total call cost c , and worst call cost $c_{\max}$, its score is $3(1 + e)/(2 + n) + .02(c + c_{\max})/(1 + n)$. Thus unobserved candidates also have defined scores. No policy obtains unpurchased counterfactual responses.

Reset retains current-round counts; cumulative memory retains all acquired counts; Window 8 retains the current round and eight preceding rounds. Observation policies and memory rules are assigned research treatments and remain fixed. Each selected arrangement changes operational visibility, aggregation, or authority under the assigned procedure. Study 2 does not implement an outer search over observation policies.

E.3 Evaluation, reporting, and validation

The six learning environments in table 11 isolate copying, private dependence, source asymmetry, and relative actor ability. The copying shift occurs before acquisition in round 25. Each environment crosses four observation policies and three memories, plus two fixed baselines, with 128 replicate trajectories of 48 rounds: $6(4 \times 3 + 2)128 = 10,752$ trajectories. Fixed baselines use an agent proposer and three agent reviewers, either exposed or blind, and buy no learning evidence. These baselines were specified before outcomes were inspected. The model-based learner selects the arrangement minimizing estimated deployment loss $3e + .02c$, where e is final error probability and c is expected production call cost. Ties use the filtered catalog order. Selection regret compares this loss with the feasible catalog minimum under the true environment parameters; an optimal-choice hit accepts every loss within 10^{-10} of

Table 11: Study 2 environments. In the environment and policy names, echo means copying. All other costs and resource ceilings are held fixed as specified in Appendix E. Numerical parameters are experimental settings, not estimates of human or language-model behavior.

Environment	q_H	q_A	ρ_A	η_H	η_A	Change
No echo	.2	.1	.1	0	0	None
High echo	.2	.1	.1	.8	.8	None
Both dependences	.2	.1	.8	.8	.8	None
Echo shift	.2	.1	.1	0	0	Both η_s become .8 at round 25
Source asymmetry	.2	.1	.2	.9	.1	None
Stronger humans	.1	.2	.2	.8	.8	None

that minimum. This objective differs from the error-first grid search and excludes acquisition and switching costs. Net value additionally charges those costs:

$$V_t^{\text{actor}} = 1 - 3e_t - .02c_t - \frac{A_t + .5S_t}{4096}, \quad (12)$$

where A_t is purchased-observation cost and S_t indicates a change from the previous arrangement. All conditions start from the exposed agent-proposer/three-reviewer arrangement; the fixed blind baseline therefore pays one initial switch. The scorer integrates production error and cost exactly. The denominator 4096 is an amortization scale for the round’s production volume, not 4096 additionally sampled production outcomes. Values from this resource schedule are not compared numerically with those from Study 1.

Learning uses root seed 2026100204, keyed by environment and round, with independent replicate rows and common potential-task panels across policies. Only purchased entries are revealed. Full-horizon and final-16-round summaries average within a trajectory before computing means and $1.96s/\sqrt{128}$ Monte Carlo half-widths across trajectories. Paired comparisons use within-replicate differences. Intervals describe randomness in acquired evidence and resulting choices, conditional on the model. The fixed baselines have deterministic expected-outcome scores.

Table 12 reports all six environments under cumulative memory; Supplementary Table S5 gives all 84 policy–memory–environment conditions, late outcomes, costs, regret, and paired comparisons. Error and net value are reported together because better selection need not repay continuing acquisition costs.

Table 12: Study 2: selection error and net value under paid observation. All six environments are shown; learning policies use cumulative memory. Entries are means $\pm$ 95% Monte Carlo half-widths over 128 trajectories of 48 rounds. Fixed baselines have exact expected-outcome scores under the model. Error is a percentage; hit is the percentage of rounds attaining a deployment-loss minimum within 10^{-10} .

Environment	Observation / fixed rule	Error (%)	Net value	Hit (%)
No echo	Paired	1.772 ± 0.002	0.86475 ± 0.00008	98.70
	Randomized	1.772 ± 0.003	0.86333 ± 0.00026	56.67
	Omit echo	1.772 ± 0.002	0.86480 ± 0.00008	99.97
	Final only	4.919 ± 0.197	0.78045 ± 0.00563	14.47
	Fixed exposed	3.520	0.87440	0.00
	Fixed blind	3.520	0.87140	0.00
High echo	Paired	1.773 ± 0.003	0.86077 ± 0.00009	99.95
	Randomized	1.782 ± 0.008	0.86078 ± 0.00024	99.82
	Omit echo	9.825 ± 0.004	0.62323 ± 0.00013	0.00
	Final only	5.983 ± 0.217	0.74760 ± 0.00640	10.89
	Fixed exposed	9.637	0.69089	0.00
	Fixed blind	3.520	0.87140	0.00
Both dependences	Paired	7.423 ± 0.049	0.69099 ± 0.00105	82.29
	Randomized	7.498 ± 0.063	0.68945 ± 0.00135	77.10
	Omit echo	15.939 ± 0.317	0.43965 ± 0.00969	0.00
	Final only	9.728 ± 0.136	0.63630 ± 0.00404	5.49
	Fixed exposed	9.919	0.68242	0.00
	Fixed blind	8.560	0.72020	0.00

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Table 12 (continued)

Environment	Observation / fixed rule	Error (%)	Net value	Hit (%)
Echo shift	Paired	1.779 ± 0.006	0.86255 ± 0.00018	98.54
	Randomized	2.113 ± 0.083	0.85206 ± 0.00238	72.40
	Omit echo	5.797 ± 0.002	0.74404 ± 0.00008	49.97
	Final only	5.648 ± 0.217	0.75770 ± 0.00621	11.20
	Fixed exposed	6.579	0.78264	0.00
	Fixed blind	3.520	0.87140	0.00
Source asymmetry	Paired	2.693 ± 0.004	0.83318 ± 0.00013	99.76
	Randomized	2.936 ± 0.046	0.82728 ± 0.00116	70.91
	Omit echo	3.546 ± 0.015	0.81160 ± 0.00044	0.00
	Final only	6.352 ± 0.191	0.73785 ± 0.00535	5.34
	Fixed exposed	5.101	0.82697	0.00
	Fixed blind	4.240	0.84980	0.00
Stronger humans	Paired	4.623 ± 0.009	0.77306 ± 0.00028	99.37
	Randomized	4.639 ± 0.014	0.77286 ± 0.00041	98.96
	Omit echo	8.969 ± 0.024	0.64781 ± 0.00072	0.00
	Final only	9.522 ± 0.263	0.63271 ± 0.00748	10.04
	Fixed exposed	19.570	0.39290	0.00
	Fixed blind	12.320	0.60740	0.00

Across environments, paired observation with cumulative memory lowers error relative to the fixed blind baseline in all six, but recovers its acquisition cost only in Stronger humans; see Section 5.4.

An independent bit-level enumeration checks 144 arrangement–environment cases to maximum absolute difference 1.11×10^{-15} . A second implementation simulates 150,000 tasks in each of four independent environments and checks 344 arrangements on the shared panels; each metric must agree within seven estimated standard errors plus 2×10^{-5} . The separate validation seed is 2026100205. Further checks cover the memory lookback boundary, copying endpoints, both resource ceilings, and the net-value identity. Exact enumeration results have no sampling interval; the Monte Carlo comparison checks implementation agreement rather than empirical validity.